Wenxin Luo

Research Interests: Large Language Models, Machine Learning, AI Security

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EDUCATION

City University of Hong Kong Jan 2024–Jun 2025
Master of Science in Data Science
GPA: 3.46 / 4.3

Beijing University of Technology (Project 211) Sep 2019–Jul 2023
Bachelor of Engineering in Software Engineering
Average Score: 85.25 / 100

PUBLICATIONS

* Equal contribution.

[1] **Wenxin Luo***, Yanzhao Chen*, Ruoxuan Cao, Cuifeng Gao, Ning Liu, Shuai Wang, Zhiyuan Wan, Daoyuan Wu. *GuardAlign: Guardrail-Aware Safety Alignment for Modern LLM Pipelines*. (Under review at ICLR 2027.) Paper Code

[2] **Wenxin Luo***, Weirui Wang*, Xiaopeng Li*, Weibo Zhou, Pengyue Jia, Xiangyu Zhao. *TAPO: Task-Referenced Adaptation for Prompt Optimization*. 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing. **(EI, CCF-B)** Paper Code

[3] **Wenxin Luo**. *Particle Swarm Optimization Algorithm with Proportional Factor Based on Nash Equilibrium*. 2022 5th International Conference on Advanced Algorithms and Control Engineering. **(EI)** Paper

RESEARCH EXPERIENCE

Lingnan University Jan 2026–Present
Research Assistant Supervisor: Prof. Daoyuan Wu

- Conducted a comprehensive literature review on the security and alignment of large language models, identifying key challenges and potential solutions.
- Developed a novel framework for evaluating the robustness of LLMs against adversarial attacks, contributing to the understanding of model vulnerabilities.
- Implemented and tested various defense mechanisms to enhance the security of LLMs.
- Developed safety-aware fine-tuning methods to improve LLM robustness against jailbreak attacks while preserving utility on benign tasks.

City University of Hong Kong Oct 2025–Dec 2025
Research Assistant Supervisors: Prof. Ning Liu, Prof. Daoyuan Wu

- Developed and maintained an academic website supporting seminar outreach and research dissemination.

Hong Kong University of Science and Technology Jun 2025–Sep 2025
Research Assistant Supervisor: Prof. Daoyuan Wu

- Explored methodologies for constructing and evaluating audio-based telecom fraud detection systems, with a focus on near-domain negative samples and LLM-driven red-teaming pipelines.

City University of Hong Kong Jan 2024–May 2025
Student Researcher Supervisor: Prof. Xiangyu Zhao

- Developed a reinforcement learning method for task-adaptive prompt optimization and designed task-referenced evaluation metrics; the resulting paper was published at ICASSP 2025.
- Built a text classifier using CNN encoding and GCN feature propagation; The dissertation received a graded A.

SKILLS & SERVICES

Languages: English (IELTS 6.5); Mandarin (Native)
Programming: Python, Java, HTML/CSS
Developer Tools: VS Code, Eclipse, Git, Linux, SSH
Libraries/Frameworks: PyTorch, Transformers, NumPy, Pandas, TensorFlow
Conference Sub-reviewer: NDSS 2026, ICML 2026, USENIX Security 2026, ICME 2026, CCS 2025